

HANDOVER
NOTES

Project Helix - Cypress Suite and QA Process Documentation
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IMPORTANT: This document covers stages 1 to 3 of the ten-step workflow only. Stages 4 to 10 are NOT DOCUMENTED here. You will need to investigate them yourself. I have left markers where I know there are gaps. Sorry - I ran out of time.

1. Why I am Writing This

I am leaving GenomeBridge at the end of this week. Daniel asked me to write up what I know about the Cypress suite and the testing process so the new QE (you) is not starting from zero. I have done my best but this is not a finished document. Some bits I did not have time to write up properly and I have marked those clearly.

The honest summary: the Cypress suite works most of the time but has problems. The selectors are too fragile, there is shared state between tests, and some of the waits are just **cy.wait(3000)** which I am not proud of. The suite passes locally more often than it passes in CI and I never got to the bottom of why.

2. Who to Talk To

Name	Role	What They Know	How to Reach Them
Daniel Brooks	Engineering Manager	Everything about what the project needs to achieve	Slack; @DanielBrooks
Aisha Rahman	Software Engineer	How the FHIR API actually works; what the data fields are	Slack; @AishaRahman
Laura Okafor	Information Governance	What data you are and are not allowed to use	Slack; @LauraOkafor
Dr Maya Patel	Product Manager	Requirements; what the ten workflow steps are	Slack; @DrMayaPatel
James Wright	Service Support	What users are actually complaining about	Email; james.wright@genomebridge.fictional
Sofia Chen	Platform Engineer	GitHub Actions; secrets management; do not touch this	Slack; @SofiaChen

3. The Workflow - What I Have Documented (Stages 1 to 3 Only)

I have only documented stages 1, 2 and 3. Stages 4 to 10 are NOT DOCUMENTED in this file. You will need to explore the system yourself and talk to Aisha and Maya.

Step	What It Does	System	My Notes	Coverage
1	Locate or register a patient	OpenMRS browser + HAPI	Ed is admin/Admin123 on the demo. Search for patient	Cypress: patient search tests
2	Review patient demographic information	OpenMRS browser	The demographic view page loads fine. Editing fails	Cypress: No navigation tests yet
3	Create a clinical referral	HAPI FHIR API (ServiceRequest)	Use Postman for this manually. The ServiceRequest is created	Cypress: No tests for this

3.1 Stages I Did Not Document

NOT DOCUMENTED: Stages 4 to 10 - consent recording, genomic test request creation, specimen association, status tracking, diagnostic report, access control, audit log. I know these exist in the FHIR API (Consent, Specimen, Task, DiagnosticReport, AuditEvent) but I did not have time to test or document them. CHECK THIS before release.

One thing I noticed but did not investigate: there is something odd about how specimens get associated with service requests. I saw some cases in the operational report where specimens appeared to be created but not linked. I do not know if this is a UI problem, an API problem or a data problem. Aisha would know more. **CHECK THIS - it might be important.**

4. The Cypress Suite - What Exists and What Is Wrong With It

The suite lives in **cypress/e2e/**. I will be honest: it is not in great shape. It was built quickly and I never got time to refactor it properly. Here is what exists and what I know is wrong.

4.1 What Exists

File	What It Tests	Status	Known Issues
authentication/login.cy.ts	Admin login, invalid login	Mostly works	Uses cy.wait(2000) before asserting login success. Fragile on slow
patient-search/search.cy.ts	Basic patient search by name	Flaky	Hardcoded search term 'Smith' - fails when demo data resets. Sel
patient-registration/registration.cy.ts	Open registration form	Incomplete	Only checks form opens. Does not test field validation or submiss
navigation/record-nav.cy.ts	Navigate to patient record from search	Works	Depends on 'Smith' search test passing first. Order-dependent. D
validation/form-errors.cy.ts	Some form error states	Partial	Only tests one error message. The selector for the error element

4.2 Known Problems in the Suite

- cy.wait() with hardcoded milliseconds instead of waiting for network requests - causes failures when the demo environment is slow
- Tests share a login state but do not always set it up cleanly - if test 1 fails at login, tests 2-5 also fail even if they would have passed
- Patient search uses the fixed string 'Smith' - the demo environment periodically resets and 'Smith' may not exist
- record-nav.cy.ts depends on search.cy.ts running first and passing - order-dependent tests are a problem when running in parallel
- CSS selectors are too specific - any OpenMRS UI update breaks them even when the feature still works
- No custom commands - everything is repeated inline in each test file
- No fixtures - test data is hardcoded in the test files themselves
- No network interception - we are not intercepting any requests to validate responses
- Screenshots are configured but not consistently captured on failure
- The CI workflow runs but takes 4-6 minutes and fails intermittently - I think this is mostly the public environment but I am not certain

4.3 Environment Configuration

Base URL	https://demo.openmrs.org/openmrs (set in cypress.env.json - do not commit this file)
Login	admin / Admin123 (public demo credentials - safe to use)
Node version	I used Node 18 LTS. Should work on Node 20 too.
cypress.env.json	In .gitignore. See .env.example for the variables needed. Ask Sofia if CI secrets are needed.

4.4 What I Would Fix If I Had More Time

- Replace all cy.wait(N) with cy.intercept() waiting for the right network request
- Build a loginAsAdmin custom command so login setup is in one place
- Replace the hardcoded 'Smith' search with a dynamically-created synthetic patient

- Break the order dependency between navigation and search tests
- Replace fragile CSS selectors with data-testid attributes or more stable alternatives
- Add a framework allocation review - some of these tests probably belong in Playwright not Cypress
- Write proper registration field validation tests using BVA on the date of birth field
- Add network interception to at least one test to validate API responses

5. Postman Collection - Partial

NOT COMPLETE: My Postman collection only covers Patient and ServiceRequest. It does not cover Consent, Specimen, Task, DiagnosticReport or AuditEvent. The collection is in api-testing/postman/ but it is a starting point only.

The HAPI FHIR server I used is <https://hapi.fhir.org/baseR4>. It is a public test server. It resets periodically so do not rely on any resource IDs I have in the collection - you will need to create your own. I stored the patient ID in a Postman environment variable called **patientId** but I forgot to export the environment file. You will need to create it yourself.

6. Open Questions I Did Not Answer

These are things I know I did not resolve. You will need to investigate them.

OQ-001: What exactly causes specimen linkage failures?

I saw them in the operational report (SPEC_UNLINK) but never confirmed whether the problem is in the browser UI, the API call, or the data. Aisha might know. CHECK THIS - it might be the most important thing to investigate.

OQ-002: Is the form validation consistent across all browsers?

I only tested in Chrome. The form validation error messages seemed to differ between Chrome and Firefox in one case but I did not have time to investigate.

OQ-003: What does the audit log look like when a specimen is created?

AuditEvent is in the FHIR spec but I do not know if the demo server supports it or what fields it returns. DEPRECATED? - needs investigation.

OQ-004: How does the session timeout actually work?

Users complain about session timeouts mid-workflow but I could not reproduce it reliably. It might be demo environment behaviour rather than a real product bug.

OQ-005: What are the actual required fields for patient registration?

The form marks some fields required but I could not tell from the source what the server actually enforces vs what is just client-side validation. This matters for the boundary value analysis tests.

OQ-006: Why does the CI suite take 4-6 minutes?

It should be faster. I never profiled it. Could be the public environment, could be test isolation issues, could be the GitHub Actions runner.

7. Data Files

Aisha has the operational queue report. Ask her for the current export. The column names use internal shorthand - there is a glossary in the company brief. The SPEC_UNLINK column is the one that seemed most relevant to the

complaints James Wright has been logging, but I never did a proper analysis.

Important: Do not use real patient data. Everything must be synthetic. Laura Okafor is very clear on this. The Synthea dataset is a good source for realistic synthetic healthcare records.

8. Miscellaneous Notes

GitHub	Repository is at github.com/alain-Sortnext/project-helix-quality-engineering . Main branch is protected. Create a feature branch for your phase work and open a PR.
Jira	Project key is HELIX. Use the Bug and Quality Risk issue types. Daniel reviews the board in standup each morning.
CI/CD	GitHub Actions. The workflows are in <code>.github/workflows/</code> . The PR check workflow runs on every pull request. Ask Sofia if you need secrets configured.
Public system rules	Do not run high-volume tests against OpenMRS or HAPI FHIR. They are shared public demo servers. Be responsible.
Credentials	Never commit credentials to GitHub. Use <code>.env</code> files locally and GitHub Secrets for CI. Sofia will be annoyed if you get this wrong.

Good luck. The project is more interesting than it looks from these notes. The specimen linkage problem in particular - I think there is something real there but I never had time to get to the bottom of it. If you do find the root cause, please tell me, I am curious.

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